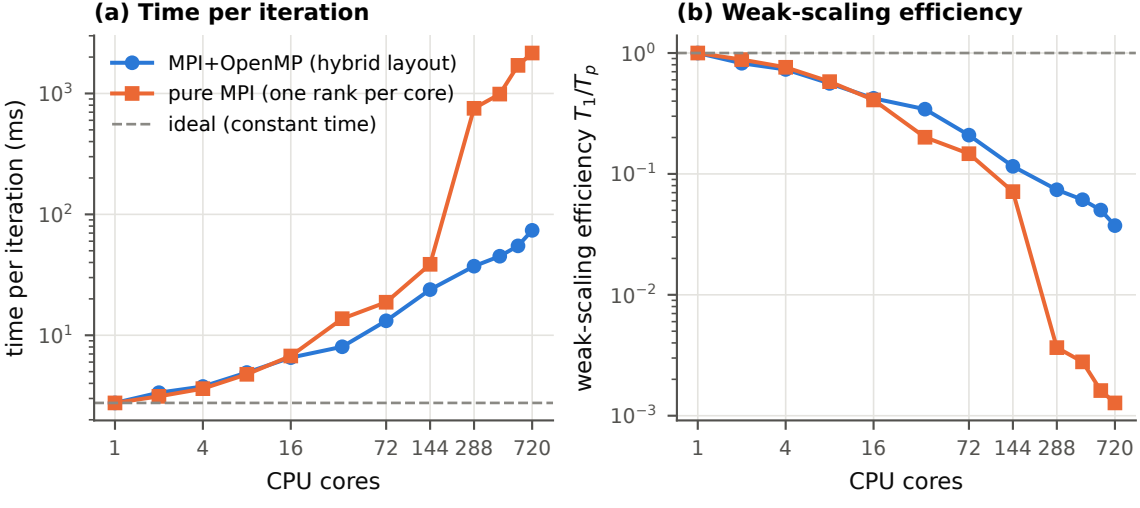
(a) Time per iteration
MPI+OpenMP (hybrid layout)
pure MPI (one rank per core)
ideal (constant time)
time per iteration (ms)
10^3
10^2
10^1
1
4
16
72
144
288
720
CPU cores
(b) Weak-scaling efficiency
weak-scaling efficiency T_1/T_p
10^0
10^{-1}
10^{-2}
10^{-3}
1
4
16
72
144
288
720
CPU cores